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SHORT-PAPER

Advancing Child Wellbeing Assessment with AI and Robotics across At-Risk Populations

JACQUELINE BORGSTEDT, Swiss Federal Institute of Technology, Zurich, Zurich, ZH, Switzerland

RICARDA WULLENKORD, Bielefeld University, Bielefeld, Nordrhein-Westfalen, Germany

FETHIYE IRMAK DOĞAN, University of Cambridge, Cambridge, Cambridgeshire, U.K.

ALVA MARKELIUS, University of Cambridge, Cambridge, Cambridgeshire, U.K.

YUE LOU, Uppsala University, Uppsala, Uppsala, Sweden

EMMA GEIJER-SIMPSON, Uppsala University, Uppsala, Uppsala, Sweden

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Jacqueline Borgstedt
ETH Zurich
Zurich Switzerland
jacqueline.borgstedt@gess.ethz.ch

Ricarda Wullenkord
Bielefeld University
Bielefeld Germany
rwullenk@cit-ec.uni-bielefeld.de

Fethiye Irmak Doğan
University of Cambridge
Cambridge United Kingdom
fid21@cam.ac.uk

Alva Markelius
University of Cambridge
Cambridge United Kingdom
ajkm4@cam.ac.uk

Yue Lou
Uppsala University
Uppsala Sweden
yue.lou@it.uu.se

Emma Geijer-Simpson
Uppsala University
Uppsala Sweden
emma.geijersimpson@uu.se

Emily S. Cross
ETH Zurich
Zurich Switzerland
ecross@ethz.ch

Friederike Eyssel
Bielefeld University
Bielefeld Germany
feyssel@cit-ec.uni-bielefeld.de

Tamsin Ford
University of Cambridge
Cambridge United Kingdom
tjf52@medschl.cam.ac.uk

Jenny L. Gibson
University of Cambridge
Cambridge United Kingdom
jlg53@cam.ac.uk

Hatice Gunes
University of Cambridge
Cambridge United Kingdom
hg410@cam.ac.uk

Georgina Warner
Uppsala University
Uppsala Sweden
georgina.warner@uu.se

Ginevra Castellano
Uppsala University
Uppsala Sweden
ginevra.castellano@it.uu.se

Abstract

Mental Health challenges disproportionately affect vulnerable children, including those with Developmental Language Disorder (DLD) and with forced migration backgrounds. These groups face elevated risks of anxiety, depression, and social withdrawal. Yet, to date, their subjective wellbeing is difficult to assess. This is particularly true for traditional self-report measures which require good reading skills, linguistic competence, and sustained attention. To overcome these barriers, we propose that social robots offer a novel opportunity to create accessible, engaging, and child-friendly wellbeing assessments.

CCS Concepts

• **Applied computing** → *Health care information systems*; *Psychology*; • **Human-centered computing** → **User centered design**; **Empirical studies in HCI**; *Accessibility systems and tools*; • **Computer systems organization** → **Robotics**; • **Social and professional topics** → **Children**.



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1 Introduction

Incidences of anxiety and depression among children and adolescents have increased globally, with concerning consequences, ranging from reduced general quality of life to rises in suicidal ideation and suicidal behaviors over the past decade [23, 24, 39]. A significant proportion of children who experience poor wellbeing or mental health challenges, however, remain unidentified and do not receive professional support and treatment [15]. Consequently, they may be prevented from being able to unfold their full potential at school and in other areas of life. Not being able to catch developing wellbeing and mental health problems early makes the prevention of adverse outcomes exceedingly difficult. One potential means of addressing these systemic gaps is the implementation of

universal, school-based wellbeing assessments to be able to identify and support at-risk children early. Indeed, schools constitute the primary setting where the majority of children can be reliably accessed and assessed. Schools have thus been described as an optimal environment for the early detection of children at elevated risk for wellbeing challenges, enabling earlier intervention and more efficient referral to appropriate treatment services [5, 18, 22]. This paper introduces a child-centered framework for robot-mediated assessment of children's subjective wellbeing in school settings. Our contribution is the operationalisation of a five-dimension wellbeing model, grounded in psychological theory and centering children's own perspectives, as a design and measurement foundation for future HRI research, with explicit attention to accessibility for children with Developmental Language Disorder and forced migration backgrounds. This framing aims to guide HRI researchers toward inclusive, scalable assessment approaches.

As a school-based approach, universal assessments typically involves the administration of a set of validated instruments to the entire student body or to specific cohorts within the institution, with the aim of identifying children at risk of poor wellbeing [22]. Despite supporting evidence [40], this approach has not been widely adopted. Research on implementation barriers highlights the substantial demands placed on school psychologists. They often lack the time, resources, or infrastructure to administer the questionnaires, even in case of electronic solutions [5, 29, 37]. Furthermore, in the cases of young children, the questionnaires oftentimes have to be administered by an adult since the reading comprehension is potentially insufficient, adding to the taxing amount of work. Taken together, the feasibility and uptake of school-based assessment practices is limited, despite moderate to high levels of acceptance from staff members [20, 30]. Consequently, alternative methods and widely accessible assessment tools are called for.

One potential avenue for realizing an automated wellbeing assessment is the use of social robots. Interactions between social robots and children have been extensively examined across academic, therapeutic, and rehabilitation contexts[16, 26, 31, 42]. For instance, the deployment of social robots has been shown to constitute an effective wellbeing intervention for children generally [25], but also in particular in supporting therapists working with children diagnosed with Autism Spectrum Disorder (ASD) [7, 38]. More recently, research started to investigate the use of robots, not only for wellbeing interventions, but also for assessing wellbeing [2, 3]. In a longitudinal study, robot-mediated interactions have been found to be a valuable tool for assessing children's wellbeing [2]. Additionally, in robot-led child–robot interactions, Abbasi et al. [1] explored whether Vision Language Models (VLMs) could be used for wellbeing assessment, highlighting both their potential and concerns about bias. Taken together, these findings suggest that social robots may offer an advantageous alternative to conventional pen-and-pencil assessments. The latter demand sustained attention from children and might not be accessible to young children[27]. Bethel et al.[4] conducted an initial exploration of interview-based wellbeing assessments delivered by a social robot, focusing specifically on experiences of bullying. Their findings yielded promising preliminary evidence: children who had been bullied were more likely to disclose their experiences to a robot than during a comparable interview with a human.

Employing social robots as scalable, automated support tools could enable the assessment of a substantially larger number of children, given that clinician-administered diagnostic assessments are time-consuming, costly, and consequently limited in accessibility. An automated assessment tool capable of flagging early indications of wellbeing difficulties would facilitate the earlier identification of at-risk children, who could then be referred to clinicians for a diagnostic assessment or directed toward appropriate support from social workers, or educators, depending on the nature of their difficulties. Compared to tablet- or app-based automated assessments, social robots offer several potential advantages in school contexts: embodied social presence may encourage engagement and disclosure, adaptive turn-taking and pacing during interaction, and the ability to integrate multimodal behavioural cues (e.g., speech, gaze, affect) alongside verbal responses.

A further limitation of traditional pen-and-paper assessments, one that is less frequently addressed in the literature on school-based assessments, concerns their accessibility. Self-report measures remain the predominant method for capturing children's subjective perceptions of their wellbeing. Including children's subjective perspectives is important since previous research suggests that ignoring children's perspectives on their own wellbeing leads to bias since domains that are subjectively important to the children remain underrepresented or get neglected[13]. However, such measures require reading, comprehension, and sustained attention, thereby limiting their accessibility for children with Developmental Language Disorder (DLD), whose language impairments may hinder their ability to understand and respond accurately to questionnaire items[17]. This is particularly problematic given that children with DLD are at increased risk of developing mental health difficulties[9, 12] and are more likely to experience poorer-quality friendships, lower levels of peer acceptance, and smaller peer networks during adolescence compared to their typically developing peers [8, 11, 28].

Similarly, children with a forced migration background face heightened risks of wellbeing-related concerns due to trauma commonly associated with displacement[10, 41]. In addition, as they navigate socializing in a non-native language, many struggle to form deeper connections with peers in the host country, with language barriers often presenting a major obstacle to initiating and maintaining friendships [36, 44]. Yet friendships are particularly important for this group: both the quantity and quality of peer relationships have been shown to play a protective role, buffering the risk of adverse psychosocial and physical wellbeing outcomes among children with forced migration backgrounds [33, 41].

This presents a complex challenge: children who struggle with language comprehension and expression, either due to a developmental condition or due to migrating to a country where they do not speak the host language, are at elevated risk for wellbeing challenges. However, the measures that would be able to detect this are not fully accessible to them. Thus, there is a need for the development of alternative, more accessible methods of assessing children's wellbeing that do not primarily rely on a child's ability to comprehend and respond to questionnaire items, whether presented in written or spoken form.

Summing up, there are serious limitations of large-scale wellbeing assessment in schools, both in terms of feasibility and accessibility. In light of pioneering evidence on the potential of social robots as alternative assessment tools, there is a clear need for further research exploring different approaches to using social robots for the assessment of children's wellbeing. The strengthening of early detection and prevention programs is needed, particularly those that can be implemented at scale and address the accessibility barriers inherent in traditional assessment approaches.

Therefore, our project will explore the feasibility, design, and validity of using social robots as an automated, inclusive method for assessing children's subjective wellbeing in school settings. This project is a collaboration between Uppsala University, University of Cambridge, Bielefeld University and ETH Zurich. In this project, we pay particular attention to children in middle childhood (8-12 years old), as this is a key period of social development that lays the foundations for their transition to adolescence [19]. Peer relations during childhood and adolescence can serve as a protective factor for adverse mental health outcomes and are thus crucial for children's wellbeing [6, 32, 43]. Based on the current challenges surrounding the assessment of child wellbeing and the limited existing understanding of how robot-child interactions may reveal meaningful markers of wellbeing, the project is structured around the following core research questions:

- **RQ1. Design & Interaction:** How can a user-centered approach guide the design of social-robot interactions that generate indicators of subjective wellbeing of children in school settings?
- **RQ2. Measurement & Constructs:** What behavioural, cognitive, linguistic, and affective correlates best characterise subjective wellbeing in 8-12-year-old children, including typically developing children and those with DLD or forced migration backgrounds?
- **RQ3. Automated Assessment:** How can AI-based computational methods be developed to automatically assess children's wellbeing from child-robot interactions, and what adaptations are needed to ensure inclusivity between diverse user groups?
- **RQ4. Broader Impact & Transferability:** How can robot-derived wellbeing measures inform our understanding of the links between wellbeing, mental health, linguistic abilities, and social interaction skills, and how can these insights be translated into ethically responsible, sustainable, and scalable real-world applications?

2 Conceptualizing and Measuring Child Wellbeing

In order to draw inferences regarding the effectiveness of social robots as technology-mediated-wellbeing assessment tools, a deep-seated understanding of the constructs that underlie wellbeing and how they are related to one another is necessary. However, there is no consensus on the construct of wellbeing in the literature. For instance, the WHO refers to wellbeing as "a positive state experienced by individuals and societies. Similar to health, it is a resource for daily life and is determined by social, economic, and environmental conditions" [35, p. 10]. Wassell and Dodge[45] define

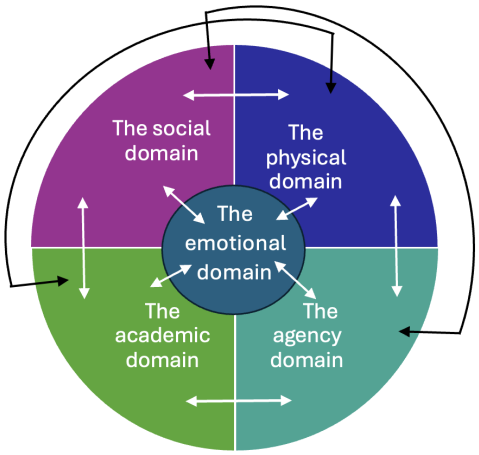


Figure 1: Project Framework for conceptualising Children's subjective Wellbeing based on work by Holder et al. [21].

wellbeing as a balance between resources and challenges. Other authors, such as [14] take a more subjective perspective and define wellbeing as "what a group or groups of people collectively agree makes a 'good life'" (p.1). This lack of consensus results in the fact that different scholars use different models of wellbeing as a basis for their respective works. To create a common ground for the HRI research within the framework of the project, we conducted a literature search. The goal was to develop a unified theoretical working model of wellbeing grounded in psychological research.

The literature search included 14 different sets of search strings and implemented four inclusion and nine exclusion criteria. Due to the high number of results regarding some of the strings (occasionally yielding more than 2 million results), we used an iterative saturation-based screening approach in which an initial minimum of 200 publications were screened for each string, which is then continued in an iterative way, where after each relevant result the next 100 publications are checked until these do not yield any relevant results. Doublets were removed in the identified relevant results, leading to a final total of 173 relevant results.

Based on these results, we propose a framework that best aligns with the goals and research questions of this project. The model takes a child-centered approach to subjective child wellbeing, i.e., takes perspectives of children on what constitutes their wellbeing into account. This approach was chosen since previous research suggests that ignoring children's perspectives on their own wellbeing leads to bias, since domains that are subjectively important to the children themselves remain underrepresented or get neglected completely[13]. Furthermore, an emphasis was put on community-related aspects of wellbeing. This was deemed especially relevant when working with children with DLD or a background of forced migration, for whom language-related barriers may influence social participation and community-related aspects of wellbeing[8, 41]. We took a positive (vs. a negative) approach to wellbeing, not understanding wellbeing solely as the absence of certain adverse factors. This is parallel to the WHO definition of health, which defines

health not as an absence of disease, but refers to how individuals perceive their physical, mental, and social functioning [34].

Holder et al.[21] was especially influential for the development of the working model, since their study provided the most in-depth consideration of children's opinions. The resulting model (Figure 1) comprises five interrelated dimensions: physical, social, academic, agency-related, and emotional wellbeing. Each dimension is conceptualized in subjective terms, focusing on the child's own evaluation rather than objective indicators, and is embedded within broader but culturally comparable contexts. The model highlights the importance of perceived social support, opportunities for self-expression and choice, satisfaction with school experiences, emotional and mental health, and the fulfillment of basic physical needs. It also incorporates methodological recommendations for measuring wellbeing in children, stressing brevity, developmental appropriateness, visual and digital support, and adaptations for children with limited language proficiency.

- The **physical** dimension: The physical dimension encompasses the current physical health state and how well physical needs of the child are taken care of, including such things as access to nutritional food, water, clothing, shelter, etc.
- The **social** dimension: The social dimension encompasses relationships with the children's family and peers. This is not based on objective markers such as the number of friends, but how well the existing relationships satisfy the child's need for relationships, and if the children receive psychological support from these relationships.
- The **academic** dimension: The academic dimension encompasses how the children do at school. This should be understood as a subjective satisfaction with different school-related areas and not be measured using objective markers such as grades. The dimension includes the relation to teachers and the presence of any adverse school experiences, such as bullying.
- The **agency** dimension: The agency dimension encompasses how much the children are allowed to have an opinion which is taken into account, are allowed to decide things for themselves within age-appropriate boundaries, and are allowed and able to have time for their own preferred activities (like hobbies, doing nothing etc.).
- The **emotional** dimension: The emotional dimension refers to emotional wellbeing, including mental health and the children's self-perception. Even though all dimensions are potentially interrelated, the emotional dimension is to be affected by and affects all other dimensions to the greatest extent.

3 User-Centred Design and Future Steps

Building on the conceptual and measurement groundwork outlined above, we propose a user-centred design approach in which focus groups with key stakeholders (caregivers, teachers, school-based counselors, other child professionals who work with the target population) are used to explore how social robots might support the assessment of wellbeing in children with DLD and a forced migration background. These discussions will address expectations, opportunities, concerns, ethical, and target group specific considerations around robot-assisted assessment, including how robots should communicate and interact with children. Thematically analysed insights will guide the design of child–robot interaction studies that sample social behaviour, elicit language, and allow to learn about children's emotional states, helping to ensure that proposed assessment scenarios are accessible and aligned with the requirements of the target child population. Given the involvement of children and the sensitive nature of wellbeing assessments, ethical, safeguarding, and data governance considerations are central to this work. These considerations will be developed in consultation with stakeholders such as schools, caregivers, parents, and ethics boards to ensure responsible and context-appropriate deployment of social robots as wellbeing assessment tools.

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